

International Trade - Empirical Project

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Method: Structure of Do File

- 1. Created unique 'Person ID' and merged blocks 4 and 5, part 1

- 2. Declared sampling pattern of the dataset using:

```
** Setting sampling pattern and multiplier
egen fs_strata = concat(State Sector Stratum Sub_Stratum_No)
lab var fs_strata "first stage strata"
svyset FSU_Serial_No [pw = Multiplier_comb], strata(fs_strata) singleunit(centered)
```

- 3. Created variables:

```
gen labor_force = .
replace labor_force=1 if Usual_Principal_Activity_Status <= 81
replace labor_force=0 if Usual_Principal_Activity_Status > 81
label variable labor_force "Employed or seeking work"

gen market_engaged=.
replace market_engaged=1 if Usual_Principal_Activity_Status <= 91
replace market_engaged=0 if Usual_Principal_Activity_Status > 91
label variable market_engaged "Employed, studying or seeking work"
```

- 4. Created three sets of age groups for different questions

```
gen working_age1 = (Age>=18 & Age<60)
label variable working_age1 "Aged 18-59"

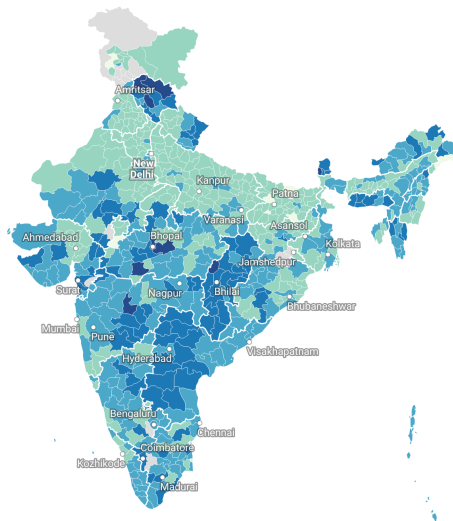
gen working_age2 = (Age>=30 & Age<=60)
label variable working_age2 "Aged 30-60"

gen working_age3 = (Age>=18 & Age<=30)
label variable working_age3 "Aged 18-30"
```

- 5. Data Wrangled using command

```
*****
**## Question a data
*****
** LFPR in India, by District (Using Usual_Principal_Activity_Status and aged 18-59)
quietly: collect: svy, subpop(working_age1): mean labor_force, over(district)
collect layout (colname) (result[_r_b _r_se _r_ci]) (cmdset)
collect clear
```

Labor force participation rates (Ages 18-59)



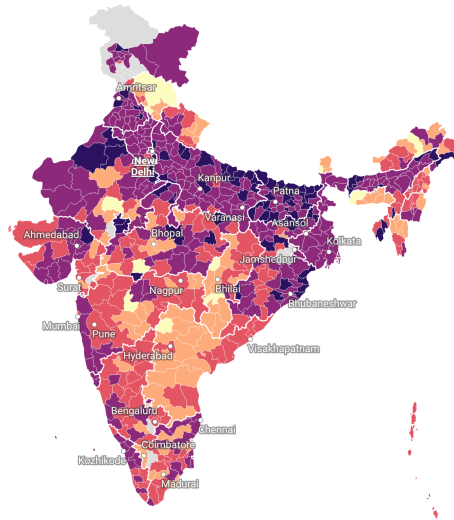
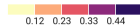
Share of working age population for which labor force = 1

Map: Lavanya, Sashank, Stuti • Source: Employment and Unemployment survey, July 2011- June 2012 (NSS 68th round) •
Map data: Datameet • Created with Datawrapper

A. Labour Force participation rates (Ages 18-59)

- **High LFPR in hilly districts** - Lahaul & Spiti (97.9%) and Alirajpur (96%) in Himachal Pradesh generally have the highest LFPR values in the dataset. Upper Siang (27%) in Arunachal Pradesh bucks the trend with the lowest LFPR in India.
- **Low LFPR across Hindi Belt** - This could reflect a combination of factors: limited non-agricultural employment, patriarchal gender norms, and out-migration of working-age men, especially in Bihar.
- **Southern and Western India** shows moderate-to-high participation. Districts in Maharashtra, Tamil Nadu, and Karnataka typically have 60% to 75% LFPR
- **Urban districts** like Mumbai (59%), New Delhi (57%), and Bengaluru (60%) actually show moderate values.

Proportion of population not engaged in market (Ages 18-59)



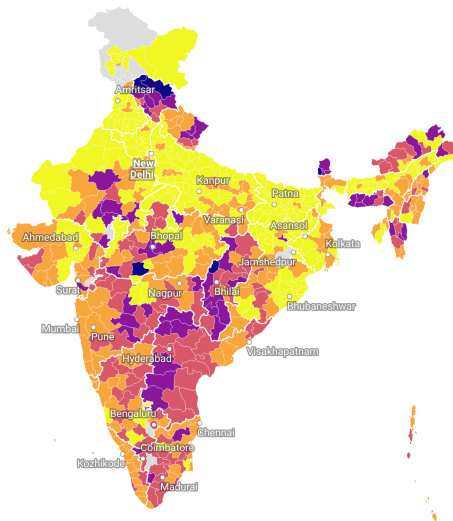
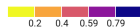
Market engagement is defined as employment, unemployed or attending educational institution. Proportion of market engagement is measured over working age population

Map: Lavanya, Sashank, Stuti • Source: Employment and Unemployment survey, July 2011- June 2012 (NSS 68th round) • Map data: Datamap • Created with Datavizmap

B. Proportion of Population not engaged in market (Ages 18-59)

- **Northern Plains top the list.** Bihar in particular – with Nawada (55.2%) and Araria (54.6%) topping the list.
- Reasons could be lack of adequate formal jobs, strong patriarchal norms, and underdeveloped markets, leading to labor underutilization.
- With a “not in market” rate of just 0.7%, almost everyone is economically active in Lahul and Spiti.
- High values in this column are usually correlated with **low female labor participation**, especially in rural Bihar, Uttar Pradesh, and parts of Rajasthan.
- **Urban centers** again show mid-range values, with cities like Delhi and Mumbai showing moderate levels (35–45%).

Female labor force participation rate (Ages 18-59)



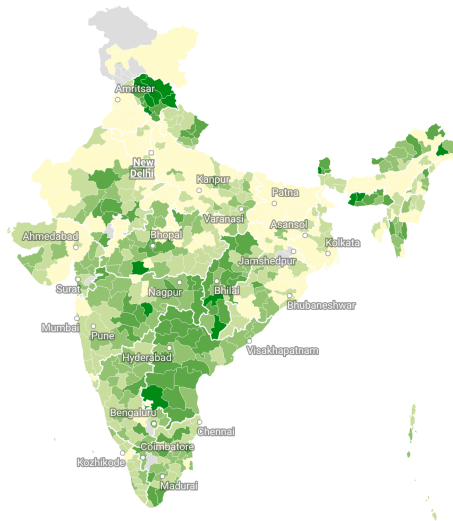
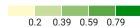
Proportion of females who are employed or unemployed and willing to work, out of total women aged 18-59

Map: Lavanya, Sashank, Stuti • Source: Employment and Unemployment survey, July 2011- June 2012 (NSS 68th round) • Map data: Datameet • Created with Datawrapper

C. Female labour force participation rate (Ages 18-59)

- **High “not in market” rates in Northern Plains** – Nawada (55.2%) and Araria (54.6%) in Bihar report the highest share of people not participating in the labor force.
- Lahaul & Spiti (0.7%) in **Himachal Pradesh** records the **lowest** “not in market” rate in the country.
- North India shows stark gender gaps – districts in Bihar, Uttar Pradesh, and parts of Rajasthan consistently display high “not in market” rates and low female LFPR, suggesting significant labor underutilization among women despite a large working-age population.
- Districts in **metros** like New Delhi (43%) and Mumbai (41%) show mid-range values for population not in the labor force.

Female labor force participation rate (Ages 30-60)

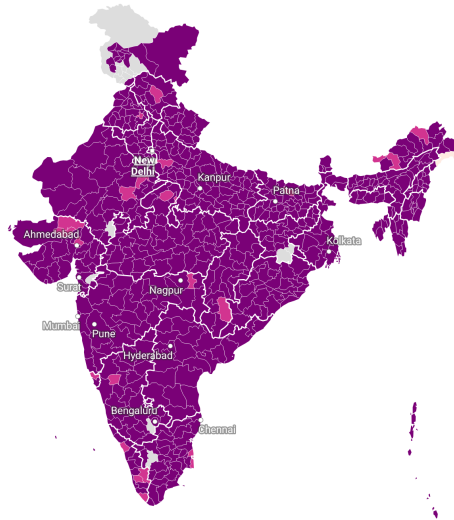
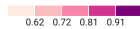


Map: Lavanya, Sashank, Stuti • Source: Employment and Unemployment survey, July 2011- June 2012 (NSS 68th round) •
Map data: Datameet • Created with Datawrapper

D. Female labour force participation rate (Ages 30-60)

- Employed women (not in education) report **highest rates in Kinnaur (82.6%) in Himachal Pradesh and West District (81.6%) in Sikkim.**
- Much of the work done by older women in this age group—especially in rural and agricultural zones—goes **unrecorded in wage or salaried data.** However, it remains visible in LFPR measures due to their roles in family enterprises, farming, and home-based work.

Male labor force participation rate (Ages 30-60)



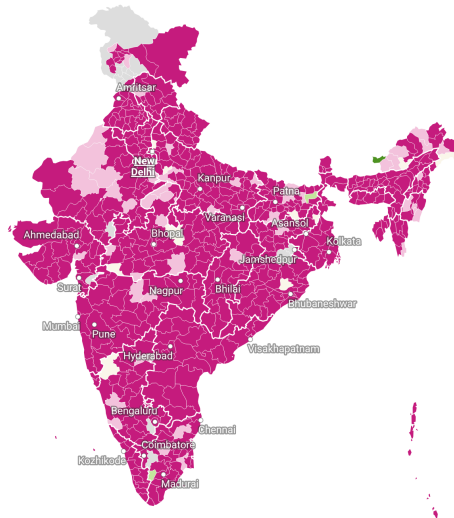
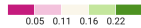
Proportion of males who are employed or unemployed and willing to work, out of total males (aged 30-60)

Map: Lavanya, Sashank, Stuti • Source: Employment and Unemployment survey, July 2011- June 2012 (NSS 68th round) •
Map data: Datameet • Created with Datawrapper

E. Male labour force participation rate (Ages 30-60)

- **Near universal male LFPR across most of India** – districts in Uttarakhand, Punjab, and Haryana report male labor force participation rates of 100% among men aged 30–60.
- Upper Siang (53.5%) in Arunachal Pradesh and Gadag (58.4%) in Karnataka show some of the lowest male LFPR values in this age group. **These exceptions may be linked to limited local opportunities, high out-migration, or underreporting of informal and seasonal work.**
- Urban districts such as Bengaluru (82%) and Kolkata (84%) report lower male LFPR than rural areas.
- **No noticeable North–South divide.**

Proportion of men not engaged in market work (Ages 18-30)



Market engagement is defined as employed, unemployed or attending educational institution. Proportion of market engagement is measured over working age population (here Ages 18-30)

Map: Lavanya, Sashank, Stuti • Source: Employment and Unemployment survey, July 2011- June 2012 (NSS 68th round) • Map data: Datameet • Created with Datawrapper

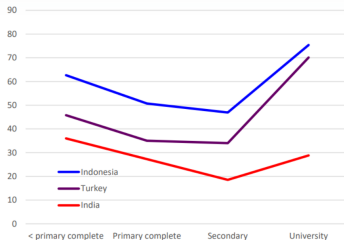
F. Proportion of men not engaged in market work (Ages 30-60)

- **Male non-participation rate across most districts is very low** – districts like Leh, Kullu in Himachal Pradesh, and Bhandara in Maharashtra report virtually no men aged 30–60 outside the labor force.
- Tawang (21.2%) in Arunachal Pradesh and Purnia (20.4%) in Bihar are outliers.

Conclusion

- India can be divided into the **Northern Plains** (Belt from Rajasthan to Assam) and the rest (**Southern States and Northeast**) based on similar patterns observed.
- The low female labour force participation rate pulls down the average labour force participation rate in the Northern Plains. Cause? U-shaped relationship with education rates.

Women's labor force participation by education, curvilinear relationships



Source: Authors' calculations from IPUMSI data.

- Better explanation– high levels of gender segregation where most white-collar jobs (that are deemed more acceptable for women in patriarchal societies) are reserved for men.

Data Sources and Reference I

- [1] E. Chatterjee and R. Vanneman. Women's low employment rates in india: Cultural and structural explanations. Working Paper No. 2021-04, 2021. Available at https://ihds.umd.edu/sites/default/files/WP2021-04_EducWLP22ndAug2021.pdf.
- [2] Ministry of Statistics and Programme Implementation. India - household consumer expenditure: Nss 68th round, schedule 1.0, july 2011 - june 2012 (type 1), September 2016. URL <http://www.icssrdataservice.in/datarepository/index.php/catalog/91>. Accessed: 2025-04-25.

Thank you